

MLB Payroll & Wins (2011-2024)

Grace Maretsky

Data Activities: Table Creating and Joining

The dataset used for this project was created by combining two primary data sources within Tableau: `mlb_standings_2011_2024.csv` and `mlb_salary_data.csv`. The standings dataset contains team performance information for each Major League Baseball season from 2011 through 2024. Key fields in these tables include season, team, wins, and other performance metrics. The second dataset contains annual payroll data for each team across the same period where salary is a main field used in this project.

To prepare the data for analysis, both tables were connected in Tableau using a common team identifier and season information. The standings dataset serves as the primary table because it contains the performance metrics that are central to the analysis. The salary dataset was then joined to this table to incorporate financial information for each team-season observation.

The join allows each team's payroll to be directly associated with its corresponding seasonal performance metrics. Structuring the data this way enables comparisons between financial investment and on-field outcomes, which is the central objective of the project. Each row in the final dataset represents a single team during a specific season, making it possible to evaluate trends across time as well as differences between teams.

Using this table structure supports the visualizations developed in Tableau, including comparisons of total payroll, payroll rank by season, payroll growth over time, win percentage trends, and the relationship between run differential and total wins. This combined dataset ultimately forms the foundation for the dashboards and story points used in the analysis.

mlb_standings_2011_2024.csv (Multiple Connecio...

Connection
☐ Live ☒ Extract ☐ Edit ☐ Refresh
 Extract contains all data: 2/2/2026 10:08:57 PM

Filters
☐ Add

mlb_standings_2011_20... mlb_salary_data

mlb_standings_2011_2024... 18 fields: 420 rows 100 rows

Name: mlb_standings_2011_2024.csv
 Description: No description available.

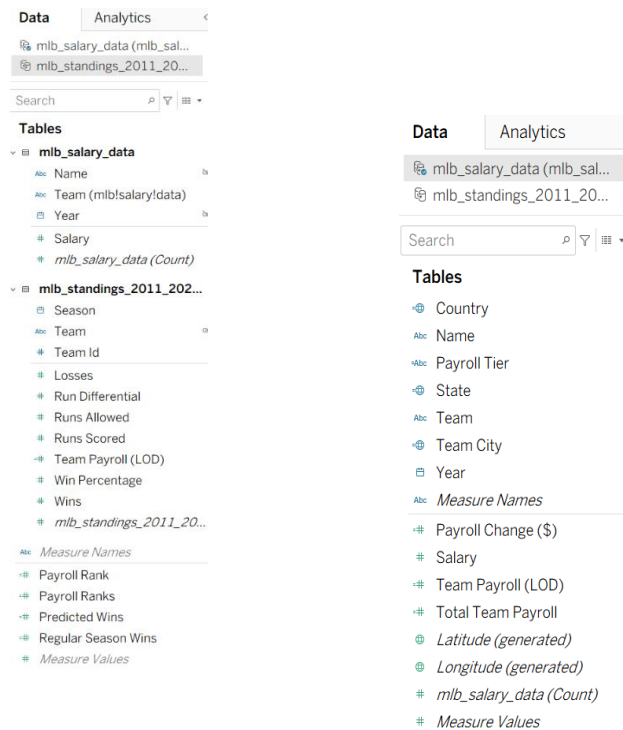
Type	Field Name	Physical Table	Rem...
Season	Season	mlb_standings_2011_2024.csv	Season
League	League	mlb_standings_2011_2024.csv	League
Division	Division	mlb_standings_2011_2024.csv	Division
Team id	Team id	mlb_standings_2011_2024.csv	Teamid
Team	Team	mlb_standings_2011_2024.csv	Team

Season	League	Division	Team id	Team
1/1/2011	null	null	158	Brewers
1/1/2011	null	null	159	O rocks
1/1/2011	null	null	163	Phillies
1/1/2011	null	null	165	Pangars
1/1/2011	null	null	116	Tigers
1/1/2011	null	null	147	Yankees
1/1/2011	null	null	138	Angels
1/1/2011	null	null	144	Braves
1/1/2011	null	null	138	Cardinals

Data Activities: Cleaning

After importing the datasets into Tableau, several data preparation and cleaning steps were performed within Tableau's data pane to ensure the data was properly structured for analysis and visualization.

First, the dataset fields were reviewed and organized within the Tableau data pane. The combined dataset includes both salary and team performance information from the mlb_salary_data and mlb_standings_2011_2024 tables. These tables contain fields related to team identification, geographic location, year of season, and performance metrics. Dimensions such as Team, Team City, Country, and State were categorized as text fields, while numerical statistics such as wins, run differential, and salary were stored as measures so that they could be aggregated and used in calculations.

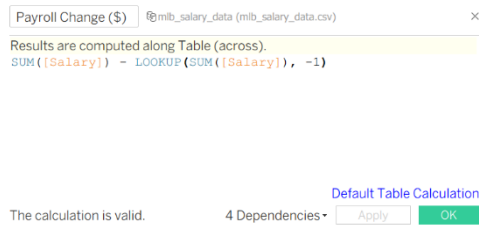


The Season and Year fields were formatted as date fields so that they could be used effectively in time-based visualizations. This formatting allows Tableau to properly interpret the data when building line charts that track payroll and team performance trends over time.

Several calculated fields were created to support the analytical components of the project. One of the primary calculated fields was Payroll Change (\$), which measures how team payroll changes from one season to the next. This was calculated using a table calculation that subtracts the previous season's payroll value from the current season's payroll value.

Calculation:

$\text{SUM}([\text{Salary}]) - \text{LOOKUP}(\text{SUM}([\text{Salary}]), -1)$

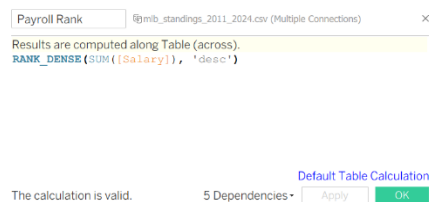


This calculation allows the analysis to identify how payroll spending increases or decreases across seasons for each team.

Another calculated field used in the project is Payroll Rank, which ranks teams based on their payroll spending. This ranking helps compare teams relative to one another rather than relying only on raw payroll values.

Calculation:

`RANK_DENSE(SUM([Salary]), 'desc')`



This calculation ranks teams from highest payroll to lowest payroll within the dataset.

Additional calculated measures such as Total Team Payroll were created to aggregate salary values and support the payroll comparisons displayed throughout the dashboards.

Calculation:

`SUM([Salary])`



This calculation is rather straightforward, yet it was a key calculation used for the Total Team Payroll KPI.

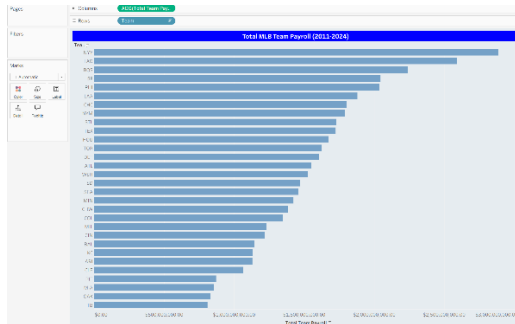
During the data preparation process, most of the unused or unnecessary fields were hidden to simplify the Tableau workspace and improve dashboard performance. Organizing the dataset in this way ensured that the visualizations and trend analyses used in the project were built on a clean and structured dataset.

Key Performance Indicators (KPIs)

The analysis in this project is guided by several key performance indicators (KPIs) that measure both financial investment and team performance. These KPIs allow the project to evaluate how payroll spending relates to competitive success in Major League Baseball between the 2011 and 2024 seasons.

Each KPI focuses on a different aspect of team spending patterns, performance trends, and competitive efficiency.

KPI #1 – Total Team Payroll



The first KPI measures total cumulative team payroll across the study period. This metric aggregates payroll spending from 2011 through 2024 to highlight long-term financial investment by each franchise.

This KPI was calculated using the Total Team Payroll measure, which aggregates salary values across all seasons for each team. The bar chart visualization allows teams to be ranked based on overall payroll spending, making it easy to identify which franchises have consistently invested the most in player salaries.

This KPI establishes the financial baseline for the analysis and highlights the significant payroll differences between high-spending teams such as the New York Yankees and Los Angeles Dodgers and lower-spending teams such as the Oakland Athletics and Tampa Bay Rays.

KPI #2 – Payroll Rank by Season



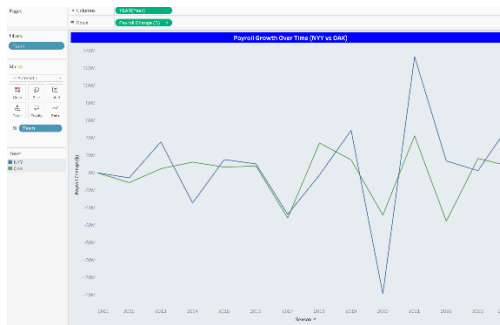
The second KPI evaluates team payroll rank within each season. Rather than focusing only on raw payroll values, ranking payroll allows teams to be compared relative to the rest of the league in a given year.

This KPI uses the previously created Payroll Rank calculated field:

`RANK_DENSE(SUM([Salary]), 'desc')`

The heatmap visualization shows payroll rank for each team across seasons from 2011 through 2024. The color scale highlights teams with consistently high payroll ranks compared to teams that operate with smaller payrolls. This KPI helps reveal long-term spending patterns and stability in team payroll strategies.

KPI #3 – Payroll Growth Over Time (NYY vs OAK)



The third KPI measures changes in payroll spending across seasons. This metric captures how team payroll fluctuates from year to year.

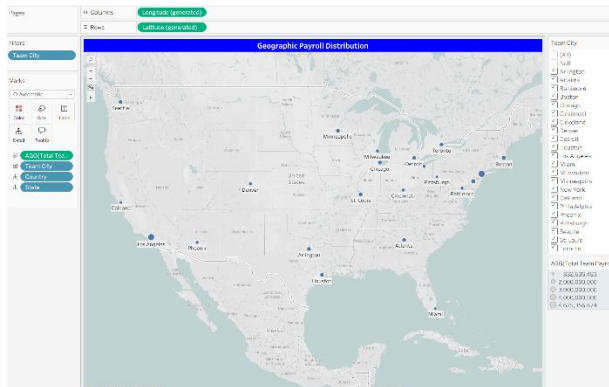
This KPI uses the calculated field Payroll Change (\$):

`SUM([Salary]) - LOOKUP(SUM([Salary]), -1)`

The visualization compares payroll growth trends for two selected teams, the New York Yankees and Oakland Athletics, which represent two very different spending approaches. The

line chart shows how payroll increases or decreases between seasons and highlights volatility in team payroll decisions. This KPI provides insight into long-term spending strategies and financial commitment.

KPI #4 – Geographic Payroll Distribution



The fourth KPI evaluates how payroll spending is distributed geographically across Major League Baseball markets.

This visualization uses Tableau's generated geographic coordinates (Latitude and Longitude) along with Team City, State, and Country fields. The size of each map marker is determined by the Total Team Payroll measure. This KPI allows viewers to observe geographic patterns in payroll spending and identify how payroll investment varies across different MLB markets.

KPI #5 – Win Percentage Over Time (8 Team Sample)



The fifth KPI evaluates team performance trends over time using win percentage. Win percentage provides a normalized measure of team success that allows comparisons across seasons.

This KPI uses the previously created Win Percentage calculated field. The line chart visualization displays performance trends across multiple teams, allowing viewers to compare how competitive performance changes over time. This KPI helps determine whether higher payroll spending corresponds to consistently stronger team performance.

KPI #6 – Run Differential vs Wins



The final KPI examines the relationship between run differential and total wins. Run differential measures the difference between runs scored and runs allowed and is often used as an indicator of overall team strength.

This KPI is visualized using a scatter plot comparing total wins and run differential, with individual seasons displayed as data points. Trend lines were added to the chart to evaluate the strength of the relationship between the two variables. This KPI provides a deeper analytical perspective by exploring whether underlying team performance metrics better explain wins than payroll spending alone.

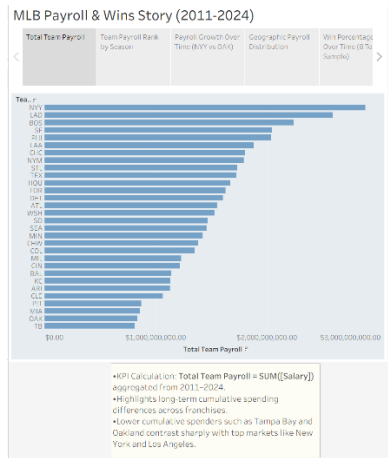
This set of KPIs establishes the analytical framework for the project. The following storyboard section builds on these metrics to interpret trends, patterns, and insights across the visualizations.

Storyboard

The top of the story is displayed below:

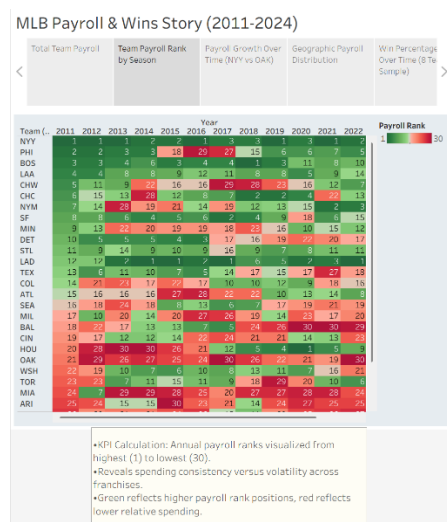


Story Page 1: Total Team Payroll



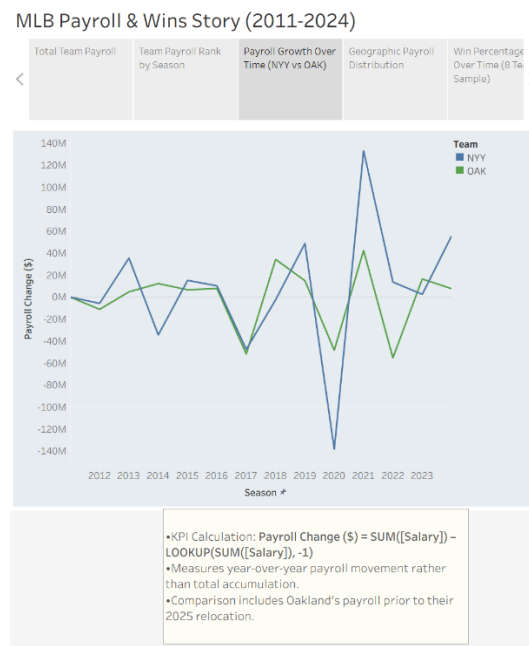
The first story page introduces total cumulative payroll spending for each Major League Baseball team between 2011 and 2024. This bar chart aggregates team salary data across the entire study period using the calculation $\text{Total Team Payroll} = \text{SUM}([\text{Salary}])$. The visualization highlights the significant differences in financial investment across franchises. Teams such as the New York Yankees and Los Angeles Dodgers appear at the top of the chart with the highest cumulative payroll spending, while smaller market teams such as the Oakland Athletics and Tampa Bay Rays appear near the bottom. This comparison establishes the financial landscape of the league and provides a foundation for examining how payroll investment varies across franchises.

Story Page 2: Team Payroll Rank by Season



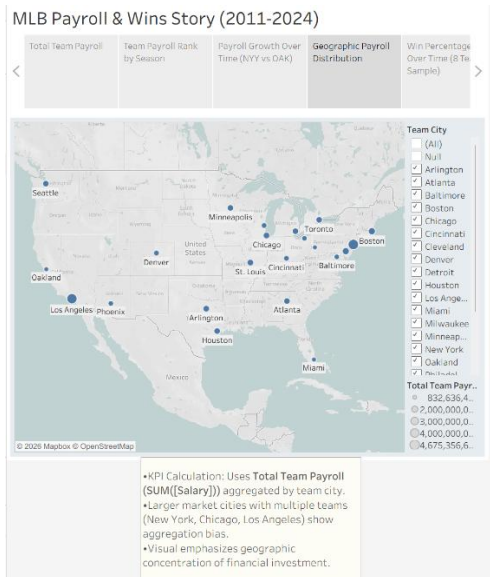
The second story page focuses on annual payroll rank across teams and seasons. A heatmap was used to visualize payroll rankings from highest spending (rank 1) to lowest spending (rank 30). The color scale allows viewers to quickly identify teams that consistently maintain high payroll rankings compared to those operating with smaller budgets. Teams such as the Yankees and Dodgers frequently appear near the top of the rankings, demonstrating sustained financial investment, while teams like Oakland and Tampa Bay consistently rank near the bottom. This visualization highlights both consistency and volatility in payroll strategies across franchises over time.

Story Page 3: Payroll Growth Over Time (NYY vs OAK)



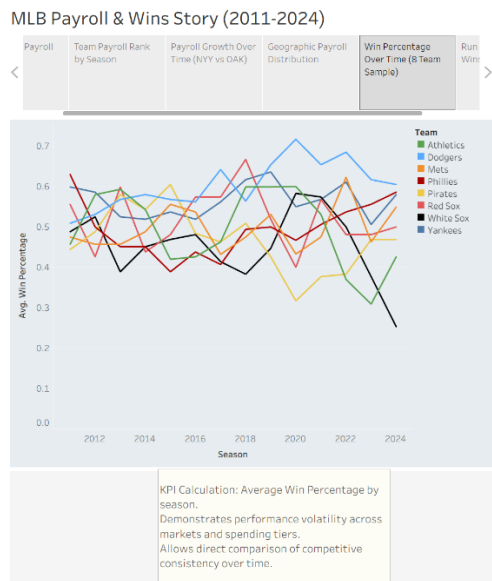
The third story page analyzes year-over-year payroll movement using the calculated field $\text{Payroll Change (\$)} = \text{SUM}([\text{Salary}]) - \text{LOOKUP}(\text{SUM}([\text{Salary}]), -1)$. Instead of cumulative spending, this visualization measures how payroll changes between seasons. The comparison between the New York Yankees and Oakland Athletics illustrates two contrasting financial strategies. The Yankees demonstrate larger payroll fluctuations and major increases in several seasons, reflecting aggressive spending patterns typical of high-revenue franchises. In contrast, the Athletics show smaller payroll adjustments and occasional decreases, highlighting the financial limitations faced by lower-revenue teams.

Story Page 4: Geographic Payroll Distribution



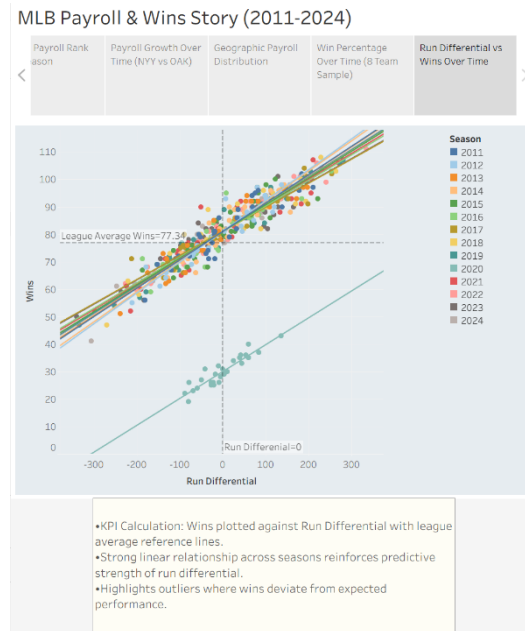
The fourth story page adds geographic context to the analysis by mapping payroll spending across MLB team locations. Using generated latitude and longitude fields, each team is plotted based on its home city, with marker size representing total team payroll. The map shows that many of the largest payroll investments occur in major metropolitan markets such as New York and Los Angeles, especially due to aggregation. Smaller markets appear with smaller markers, reflecting lower payroll spending. This visualization helps illustrate how market size and geographic concentration can influence financial investment in MLB franchises.

Story Page 5: Win Percentage Over Time (8 Team Sample)



The fifth story page transitions from financial metrics to on-field performance. A line chart displays win percentage trends for a sample of eight teams across the study period. This visualization allows viewers to compare performance volatility across teams and seasons. While some higher payroll teams (such as the Dodgers), maintain relatively consistent performance levels, the chart also demonstrates that success varies from year to year across all teams. This suggests that financial investment alone does not guarantee sustained competitive success.

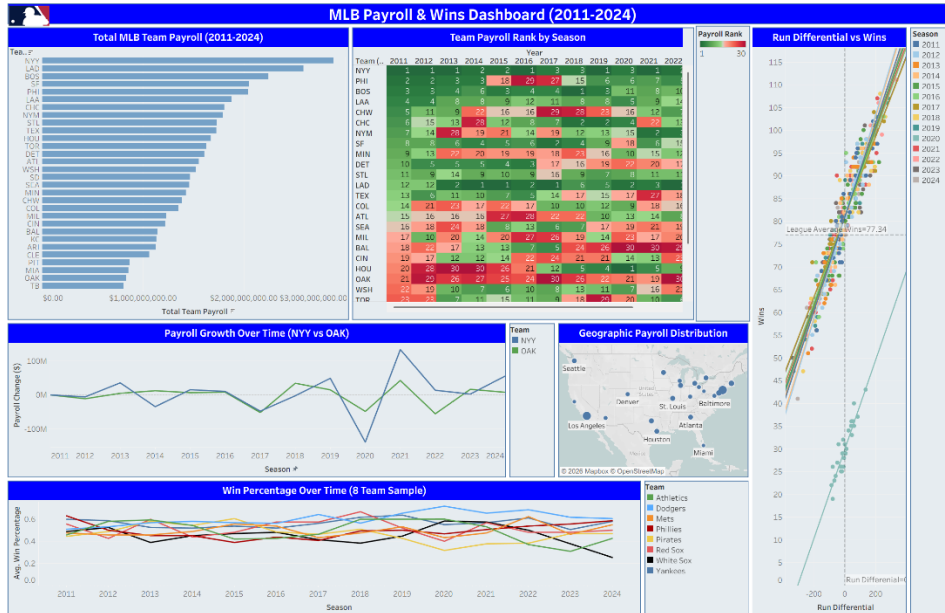
Story Page 6: Run Differential vs Wins



The final story page evaluates the relationship between run differential and total wins using a scatter plot with regression trend lines. Each point represents a team's performance in a specific season, with run differential plotted on the horizontal axis and total wins on the vertical axis. Reference lines mark league average wins and a neutral run differential. The upward slope of the trend lines indicates a strong positive relationship between run differential and wins, reinforcing the idea that teams that outscore opponents by larger margins tend to win more games. This visualization highlights run differential as a strong statistical indicator of team performance.

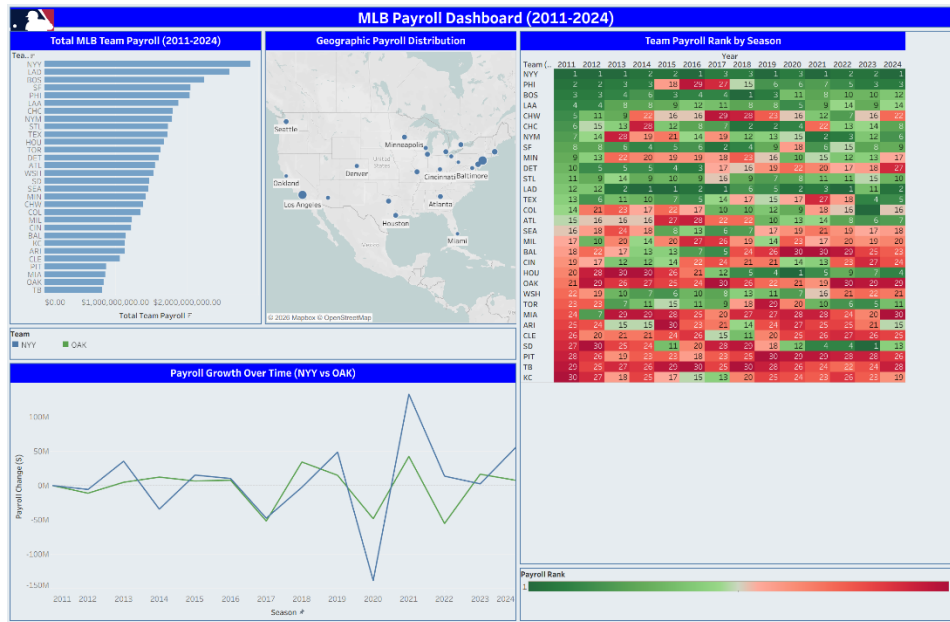
Dashboards

MLB Payroll & Wins Dashboard (2011–2024)



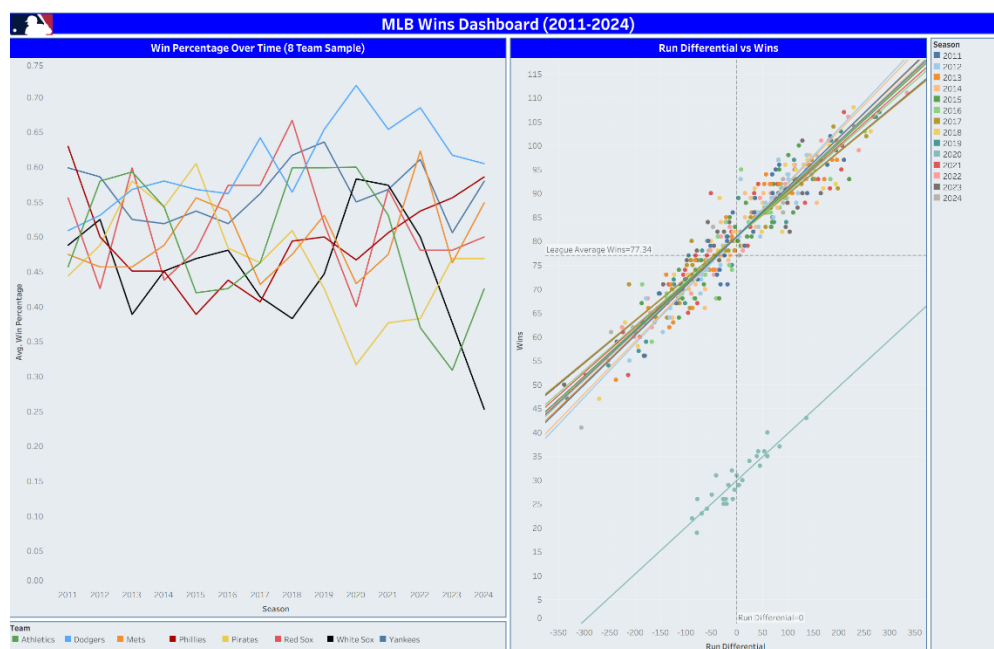
The primary point of reference for the project is the combined MLB Payroll & Wins Dashboard (2011–2024). This dashboard integrates the major financial and performance indicators presented throughout the storyboard, including total team payroll, payroll rank by season, payroll growth over time, geographic payroll distribution, win percentage trends, and the relationship between run differential and wins. By presenting these visualizations together, the dashboard allows users to analyze how payroll investment and on-field outcomes interact across teams and seasons. This combined view provides the most complete perspective of the dataset, allowing patterns between spending, performance, and competitive efficiency to be observed simultaneously.

MLB Payroll Dashboard (2011–2024)



The MLB Payroll Dashboard (2011–2024) focuses specifically on the financial aspects of the analysis. This dashboard highlights cumulative payroll spending, payroll rank by season, geographic payroll distribution, and payroll growth trends. By isolating these financial metrics, the dashboard allows viewers to better understand how payroll investment varies across franchises and over time. When compared with the combined dashboard, this view provides a more focused look at the economic structure of the league, illustrating the significant gap between high-spending franchises and low-spending franchise.

MLB Wins Dashboard (2011–2024)



The MLB Wins Dashboard (2011–2024) shifts the analysis toward on-field performance indicators. This dashboard includes the win percentage over time visualization and the run differential versus wins scatter plot, both of which evaluate competitive outcomes across seasons. These visualizations help illustrate how team performance fluctuates over time and highlight the strong relationship between run differential and total wins. When viewed alongside the combined dashboard, this performance-focused view helps contextualize whether financial investment translates into measurable success on the field.

Conclusion

This analysis highlights several important takeaways regarding the relationship between payroll investment and team performance in Major League Baseball from 2011 through 2024. The results suggest that while higher payroll spending does increase the probability of

competitive success, financial investment alone does not guarantee strong performance. Teams with larger payrolls often have more resources to build deeper rosters and acquire talent, but the visualizations show that outcomes on the field still vary across seasons. The analysis also reinforces the strong relationship between run differential and total wins, indicating that scoring efficiency and run prevention are more reliable drivers of success than payroll size alone.

Overall, the findings suggest that while spending more can improve a team's chances of winning, consistent results are ultimately determined by how effectively teams convert that investment into run production and overall on-field performance.

Tableau Public

[https://public.tableau.com/app/profile/grace.maretsky6140/viz/MLBPayrollWinsProject/
MLBPayrollWinsDashboard?publish=yes](https://public.tableau.com/app/profile/grace.maretsky6140/viz/MLBPayrollWinsProject/MLBPayrollWinsDashboard?publish=yes)